IGCSE 0607 Practice Questions — E1.11 — Combined Questions & Markscheme

Rates

Total Questions: 6 — **Total Marks:** 30 Marks

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1 A train departs at 08 : 15 and arrives at its destination at 09 : 45.
The train travels a distance of 120 km.
Calculate the average speed of the train.

..... km/h **[2]**

2 A car travels 120 km in 1 hour and 30 minutes.

Calculate the average speed of the car in km/h.

..... km/h **[3]**

3 Water flows from a pipe into a rectangular tank at a rate of 45 litres per minute.

(a) Calculate the volume of water, in cubic centimetres (cm^3), that flows into the tank in 1 second.

..... cm^3 **[2]**

(b) The tank has a base measuring 1.5 m by 80 cm. Calculate the rate at which the water level is rising in the tank, giving your answer in cm/s.

..... cm/s **[3]**

4 A car uses fuel at an average rate of 6.4 litres per 100 km.

(a) Calculate the amount of fuel used for a journey of 350 km.

..... litres **[2]**

(b) The cost of fuel is 1.35 euros per litre.
The exchange rate is \$1 = 0.85 euros. Calculate the cost of the fuel for this journey in dollars (\$).
Give your answer correct to the nearest cent.

\$ ..... **[3]**

5 A cyclist travels from Town A to Town B, a distance of 48 km.
The cyclist travels the first 30 km at an average speed of x km/h. They travel the remaining distance at an average speed of $(x - 4)$ km/h. The total time taken for the journey is 2 hours and 30 minutes.

(a) Show that $5x^2 - 116x + 240 = 0$.

[4]

(b) Solve the equation $5x^2 - 116x + 240 = 0$ to find the possible values of x.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

6 A man runs 15 km at an average speed of x km/h.
He then walks 12 km at an average speed of $(x - 4)$ km/h.
The total time taken for the journey is 3.5 hours.

(a) Show that x satisfies the equation

$$7x^2 - 82x + 120 = 0$$

[3]

(b) **(i)** Solve the equation

$$7x^2 - 82x + 120 = 0$$

$x =$ ..... , or $x =$ ..... **[3]**

(ii) Find the time taken for the man to run the 15 km.

..... hours **[2]**

1 Time taken = 1.5 hours (or 1 hour 30 minutes) **[B1]**

$120 \div 1.5 = 80$ **[B1]**

[Back to Question] *[Total Marks: 2]*

2 Time = 1.5 hours **[B1]**

Average Speed $= \dfrac{120}{1.5}$ **[M1]**

80 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 (a) 45×1000 or $\dfrac{45}{60}$ **[M1]**

$\dfrac{45000}{60} = 750$ **[A1]**

(b) Base area $= 150 \times 80 = 12000$ cm^2 **[M1]**

Rate of height increase $= \dfrac{\text{Volume per second}}{\text{Base area}} = \dfrac{750}{12000}$ **[M1]**

0.0625 **[A1]**

[Back to Question] *[Total Marks: 5]*

4 (a) $\dfrac{350}{100} \times 6.4$ **[M1]**

22.4 **[A1]**

(b) Cost in euros $= 22.4 \times 1.35 = 30.24$ **[M1]**

Cost in dollars $= 30.24 \div 0.85$ **[M1]**

35.58 **[A1]**

[Back to Question] *[Total Marks: 5]*

5 (a) $\dfrac{30}{x} + \dfrac{18}{x-4} = 2.5$ **[M1]**

$\dfrac{30(x-4)+18x}{x(x-4)} = 2.5$ **[M1]**

$48x - 120 = 2.5(x^2 - 4x)$ **[M1]**

$48x - 120 = 2.5x^2 - 10x \implies 2.5x^2 - 58x + 120 = 0 \implies 5x^2 - 116x + 240 = 0$ **[A1]**

(b) $x = \dfrac{-(-116) \pm \sqrt{(-116)^2 - 4(5)(240)}}{2(5)}$ **[M1]**

$x = \dfrac{116 \pm \sqrt{13456 - 4800}}{10} = \dfrac{116 \pm \sqrt{8656}}{10}$ **[M1]**

$x = 20.90$ or $x = 2.30$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $\dfrac{15}{x} + \dfrac{12}{x-4} = 3.5$ **[M1]**

Multiply by $2x(x-4)$ to clear denominators (or equivalent): $30(x-4) + 24x = 7x(x-4)$ **[M1]**

$54x - 120 = 7x^2 - 28x \implies 7x^2 - 82x + 120 = 0$ **[A1]**

(b) **(i)** $(7x-12)(x-10) = 0$ OR $\dfrac{82 \pm \sqrt{(-82)^2 - 4(7)(120)}}{14}$ **[M2]**

If M0, allow M1 for $(7x \pm a)(x \pm b) = 0$ where $ab = 120$ or $7b \pm a = -82$

$x = 10$, $x = \dfrac{12}{7}$ (or 1.71) **[A1]**

(ii) Identifies valid speed $x = 10$ (since $x - 4 > 0 \implies x > 4$) **[B1]**

Time $= \dfrac{15}{10} = 1.5$ **[B1]**

[Back to Question] *[Total Marks: 8]*